

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

The daily high temperatures (°F) recorded in Cedar Falls over the last twelve days are given below. Use this data set for problems 1–6.

69, 62, 73, 53, 79, 69, 66, 58, 73, 62, 83, 69

1. Find the mean.

Answer: $\text{sum } 816 \div 12 = 68$

2. Find the median.

Answer: sorted: 53, 58, 62, 62, 66, 69, 69, 69, 73, 73, 79, 83 → median $(69 + 69)/2 = 69$

3. Find the mode.

Answer: 69 (appears three times)

4. Find the range.

Answer: $83 - 53 = 30$

5. Find the lower quartile (Q1) and the upper quartile (Q3).

Answer: lower half 53–69: $Q1 = 62$; upper half 69–83: $Q3 = 73$

6. Find the interquartile range (IQR).

Answer: $IQR = 73 - 62 = 11$ (accept a correct IQR from the student's own #5)

7. Write the five-number summary of this NEW data set: 11, 16, 19, 23, 29, 35, 44

Answer: min 11, $Q1 = 16$, median 23, $Q3 = 35$, max 44

Algebra 1 – Module 12 Practice Test •

WORKED KEY

8. Identify the outlier, and explain how you know: 66, 69, 71, 74, 112

Answer: 112 — it lies far above the rest of the data

9. For the data 35, 37, 41, 43, 104, find the mean with and without the outlier 104. Describe how the outlier affects the mean.

Answer: with: $260 \div 5 = 52$; without: $156 \div 4 = 39$ — the mean DROPS from 52 to 39

12. A report lists the number of followers for each student in one class, where one student is a famous influencer. Which measure better represents a TYPICAL value — the mean or the median? Explain your reasoning.

Answer: the MEDIAN — the outlier pulls the mean up, so the median better represents a typical student

Full Name: _____

Period: _____

10. Two robotics teams competed in the same three rounds. Their round scores are shown. Show that the two teams have the SAME mean but DIFFERENT ranges.

Team	Round Scores
Team Red	68, 74, 80
Team Blue	62, 77, 83

Answer: means: Red = $222/3 = 74$, Blue = $222/3 = 74$ — EQUAL; ranges: Red = 12, Blue = 21 → Team Blue has the greater spread

11. Error Analysis — For the data 17, 26, 41, 9, 30, Marcus said the range is 41. Identify his mistake, then find the range.

Answer: he gave the maximum, not the range — range = max – min = $41 - 9 = 32$

13. Devin's first three test scores are 74, 85, and 93. What score does he need on his fourth test so that his mean is exactly 87? Show your work.

Answer: need total $87 \times 4 = 348$; $348 - 252 = 96$

Cumulative Review — ACT Style. Circle the best answer.

14. What is the vertex of $y = x^2 - 6x + 11$?

- (A) (3, 2)
- (B) (-3, 2)
- (C) (3, 11)
- (D) (6, 11)

Answer: $x = 6/2 = 3$; $y = 9 - 18 + 11 = 2$

15. Find the product: $(4x + 1)(x + 3)$

- (E) $4x^2 + 13x + 3$
- (F) $4x^2 + 12x + 3$
- (G) $4x^2 + 13x + 4$
- (H) $5x + 4$

Answer: FOIL: $4x^2 + 12x + x + 3$

16. Which ordered pair solves the system $y = x + 5$ and $x + y = 11$?

- (A) (3, 8)
- (B) (8, 3)
- (C) (3, 3)
- (D) (4, 9)

Answer: $x + (x + 5) = 11 \rightarrow x = 3, y = 8$